

1. Agenda

Content

- Basic intuition for classifier
- Simple searching algorithm
- How to solve the optimization problem
- Defining the classification problem mathematically
- Relationship between distance and gain function
- Functions
 - Limits
 - Continuity
 - Domain and range
- Some important functions

2. Basic intuition for classifier

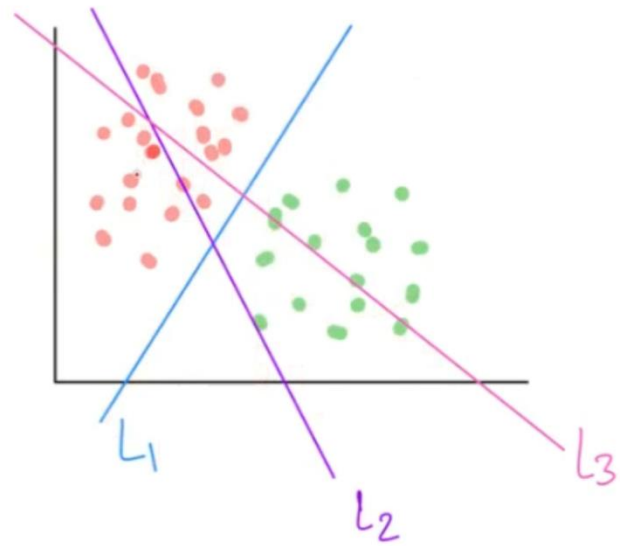
Basic intuition for classifier

We have a classification problem and our task is to find the best line that separates the 2 classes

Q. Do you agree that there are infinite number of lines that can separate the 2 classes ?

- Yes

Basic Intuition for Classification

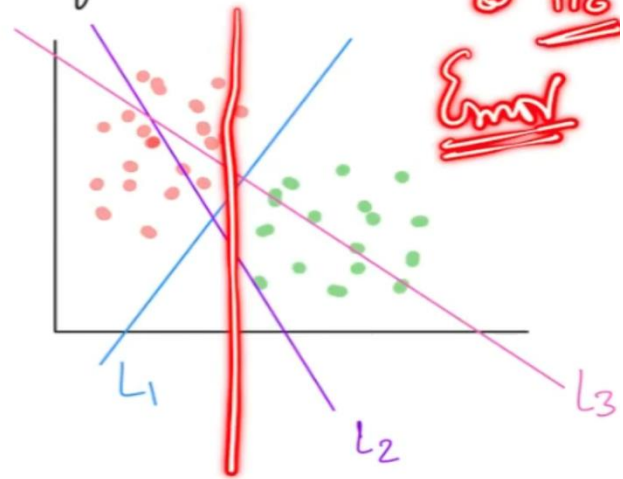


SCALER

Q. Is there a way in which we can calculate some kind of error for each line?

- Do you agree that line L1 is properly classifying all the red and green points hence we can say that it is not making any errors
 - Thus, in an ideal situation the error should be 0
- Now let say we also have another line which is misclassifying a single red point as belonging to green class rest all the other points are classified correctly
 - In this case we can define the error as Number of misclassified points divided by total number of points times 100
 - so for this line the error would be 1% if we had 100 points in total

Basic Intuition for Classification



Similarly we can calculate the error for all the lines, next what we can do is we can create a error plot of all the possible lines that will separate the 2 class points as we can see below.

Q. From all the possible lines which line should we pick by looking at the error plot?

- we should pick that line which has the lowest error

Q. By looking at the error plot can we also say that the error is following some kind of function

- yes, so the red line would be the function of error of different lines that would separate the points

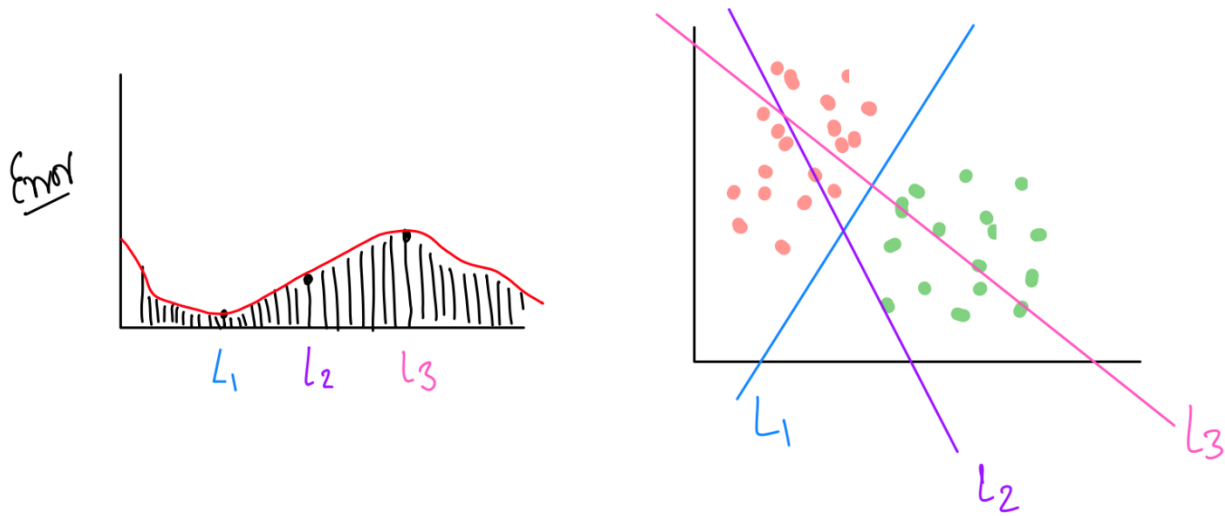
Q. In the ideal case do we want the error to be maximum or minimum?

- Minimum

Note:

- that this error function changes depending upon the algorithm we choose we'll learn about them in the upcoming modules
- This error is also called as the loss function or the cost function and we want this ideally to be 0

Basic Intuition for Classification



Q. Let say we do not know about calculus then how would we go about finding the best possible line in the above error plot?

- we would go one by one over all possible lines to find the line that has the least error
- essentially we would perform a linear search

Let's understand why this is not an optimal way to find the best possible line

- Now let's say that we vary this weight values from -10 to +10 with a step size of 0.1

Q. How many values do you iterate over for w_1 ?

- We'll have in total of 201 points between -10 to +10 with a step size of 0.1
- we'll be having similar 201 values to go over for w_2 and w_0
-

Q] Can we do optimisation using linear search?

If I am doing a linear search,

$w_1 \rightarrow [-10 \text{ to } 10]^{0.1}$ $[-10, \dots, 10]$
 $w_2 \rightarrow [-10 \text{ to } 10]^{0.1}$ (201)
 $w_0 \rightarrow [-10 \text{ to } 10]^{0.1}$ (201)

Total possibilities = 201^3

Diagram illustrating a linear search process over a range from -10 to 10 with a step of 0.1. The range is divided into segments of 100, 1, and 100, with a total of 201 steps.

```
In [3]: (201**3)*(10**-6)
```

```
Out[3]: 8.120600999999999
```

```
In [4]: (201**4)*(10**-6) # 27 mins
```

```
Out[4]: 1632.240801
```

```
In [5]: (201**5)*(10**-6) # 91 hours
```

```
Out[5]: 328080.40100099996
```

```
In [9]: ((201**6)*(10**-6))/(60*60*24) # 2 years
```

```
Out[9]: 763.2425995509375
```

```
In [11]: ((201**7)*(10**-6))/(60*60*24*365)
```

```
Out[11]: 420.3061986568176
```

How to solve optimization problem

To solve this optimization problems we'll learn about the concept called as Gradient descent

- what it does is it allows us to find/search for an approximately correct solution

To properly understand what gradient descent is we'll first need to understand below concepts

- What is maxima and minima of a function
- Calculus in multi variable systems
- Calculus in single variable systems
 - To understand calculus we'll need to understand
 - What is Derivative, slope and tangent
 - And to understand this we need a basic understanding of
 - Limits and continuity and differentiability

This is the syllabus for the upcoming 4 classes and we'll be going from the bottom to top ie learning the basics first and then the advance concepts

How to solve optimisation problems?

→ Use Gradient Descent
It allows us to find the approximately correct solution very quickly!

↳ maxima, minima

↳ Calculus in multi variable

↳ Calculus in single variable

↳ derivative, slope, tangent

↳ limit, continuity, differentiability

7. Defining the classification problem mathematically

Defining the classification problem mathematically

Let's say we have a dataset with 2 features and a target variable and it is a binary classification task so how do we represent this information mathematically?

Defining the classification problem mathematically

Given: labelled dataset

$$\mathcal{D} = \left\{ (\bar{x}_i, y_i) : \bar{x}_i \in \mathbb{R}^d, y_i \in \{-1, 1\} \right\}_{i=1}^m$$

Labels
features

Goal: Find a function $f(\bar{x})$ such that

$$f(\bar{x}) = \hat{y}_i$$

predicted value

ideally $\hat{y}_i = y_i$
predicted value *Actual Value*

f_1	f_2	T
●	●	●
●	●	●
●	●	●
●	●	●
●	●	●

Q. So now what is the next step we are passing the points from dataset into our function $f(x)$ and get the predicted value for each point next we update the weight vectors to get a final function how did we do that?

- Do you remember that we studied about the gain function where we wanted to maximise the distance of data points from the hyperplane
 - The input to the gain function is our dataset $\mathcal{D}$, W^+ W^- and w_0
- And this is the function that we want to optimize because this gain function will measure how good the classifier function is performing

Defining the classification problem mathematically

Given: labelled dataset $\mathcal{D} : \left\{ (\bar{x}_i, y_i) : \bar{x}_i \in \mathbb{R}^d, y_i \in \{-1, 1\} \right\}_{i=1}^m$

Labels (pointing to y_i)
feature (pointing to $\bar{x}_i$)

f_1	f_2	T

Goal: Find a function $f(\bar{x})$ such that

Classification

$f(\bar{x}) = \hat{y}_i$ (predicted value) ideally $\hat{y}_i = y_i$ (Actual Value)

$\text{sign}(\bar{w}^T \bar{x} + w_0)$ $\hat{y}_i$ or y_i'

predicted value (pointing to $\hat{y}_i$)
predicted value (pointing to $\hat{y}_i$)
Actual Value (pointing to y_i)

This function will measure how good classifier is performing.

$\mathcal{L}(\mathcal{D}, \bar{w}, w_0) \rightarrow$ This is a function that we want to optimise.

Relationship between distance and gain function mathematically

what is the relationship between distance & gain function?

$$\textcircled{1} \quad \mathcal{L}(\mathcal{D}, \bar{w}, w_0) = \sum_{i=1}^m g(\bar{x}_i, y_i, \bar{w}, w_0)$$

$$\textcircled{2} \quad \mathcal{L}(\mathcal{D}, \bar{w}, w_0) = \prod_{i=1}^m g(\bar{x}_i, y_i, \bar{w}, w_0)$$

Q. Which is better equation 1 or 2 or both?

- here in equation 1 we are summing all the distances of each point

- in equation 2 we are multiplying all the distances of each point

Equation 1 is the correct way of representing the gain function lets understand why

Lets consider a scenario where we have our 2 class data points clustered together

- we can now compute the distance of each point from the classifier
- what if there is a point which sits on the line what would that distance be from the line?
 - it would be 0 so now can you think what would happen when we multiply distances of all the points?
 - the final value will become 0 and this is the reason why we are summing all the distances of the points and not multiplying them

T

function?

✓

$$\textcircled{1} \quad g(D, \bar{w}, w_0) = \sum_{i=1}^n g(\bar{x}_i, y_i, \bar{w}, w_0)$$

✗

$$\textcircled{2} \quad g(D, \bar{w}, w_0) = \prod_{i=1}^n g(\bar{x}_i, y_i, \bar{w}, w_0)$$



Functions

Let's understand the basics of what is a function

when we write $f(x)=y$

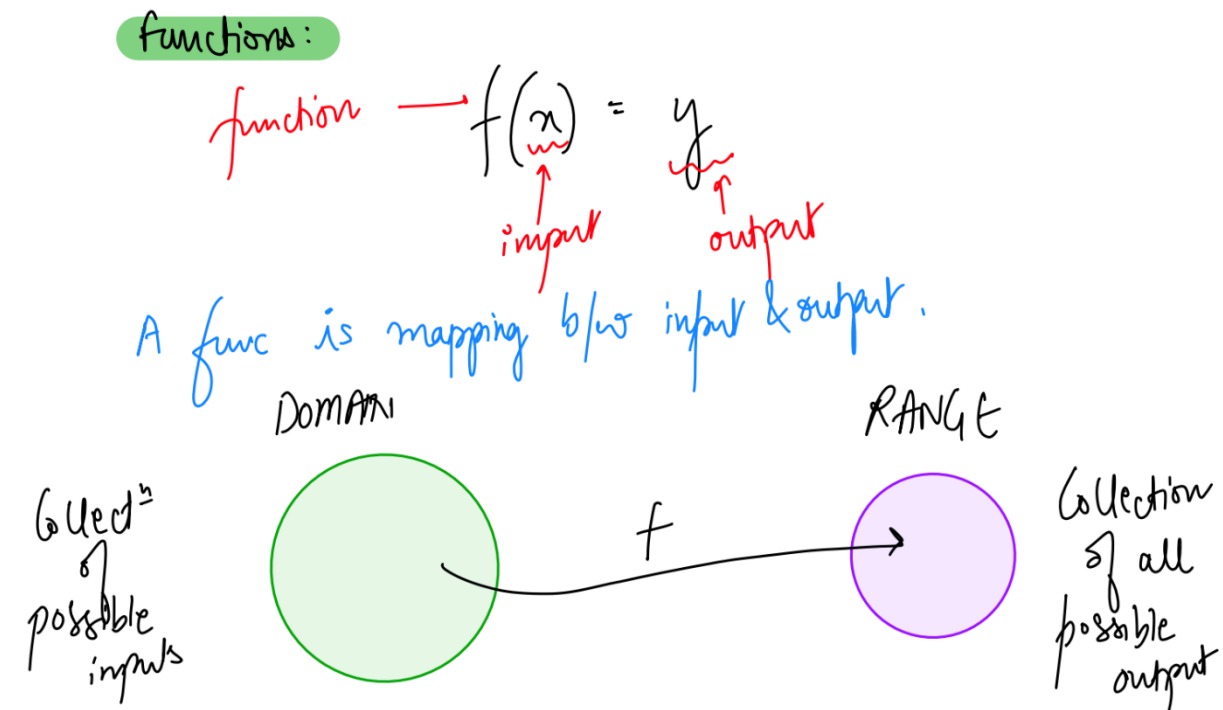
- $f(x)$ is called as a function
- and it is a mapping between the input x and the output y

Q. What do we call collection of all possible inputs as?

- We call that as **Domain**
- **Domain** are all x axis values that function is allowed to take, eg when we speak of logs we don't define it for negative numbers only positive so therefore when we speak of a function we need to define what are the values allowed or those make sense

Q. What do we call the collection of all possible outputs?

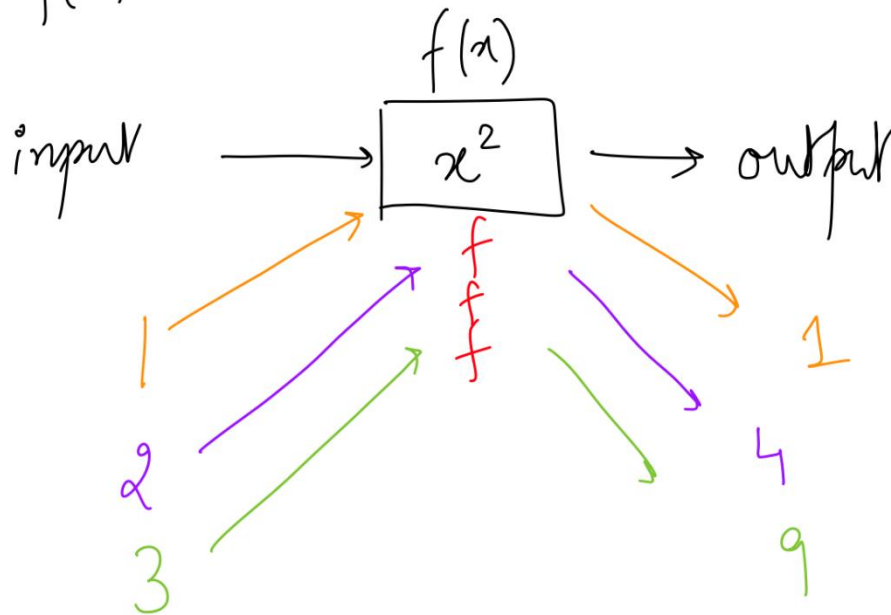
- We call it as **Range**



Let's define a function $y=f(x)=x^2$ here it means that x is mapped to y in the format of x^2

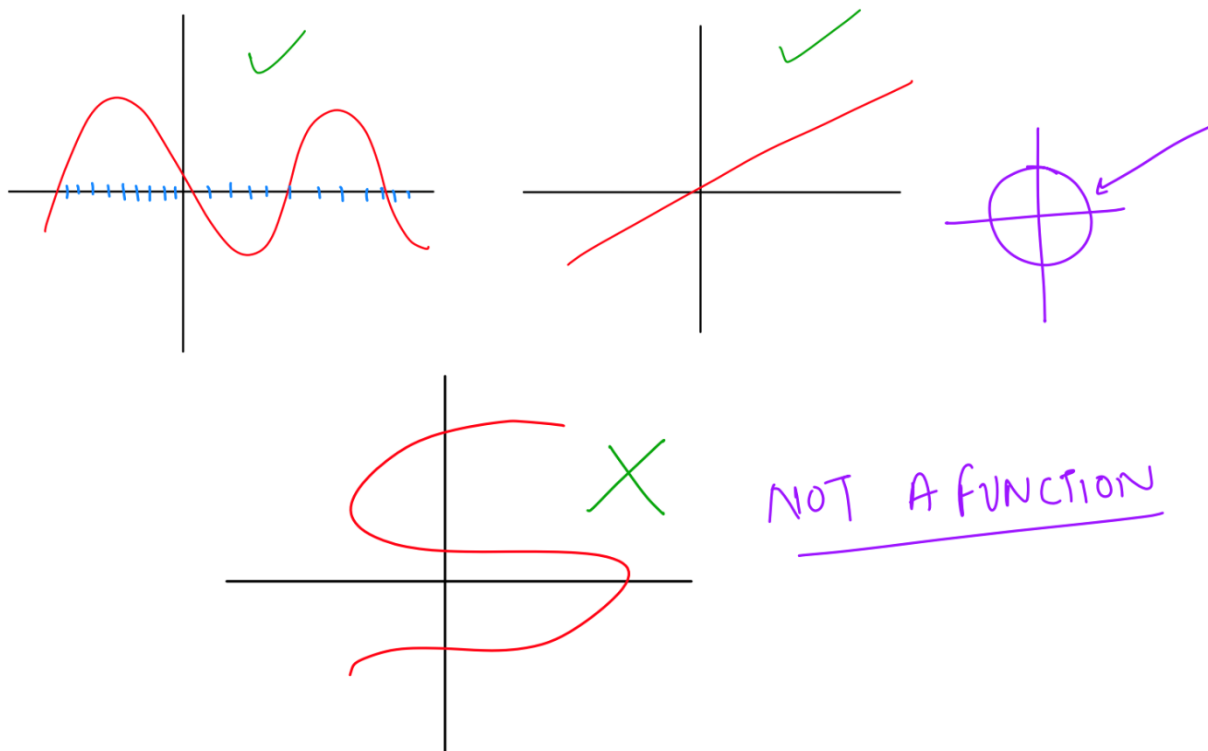
- here 1,2,3 becomes our domain and the output 1,4,9 becomes our range for the given function

$$y = f(x) = x^2$$



Instructor note:

- You can draw the functions and ask them can we classify it as a function or not



Q. How do we identify if a given function is a valid function or not?

- For a function to be identified as a function it should satisfy the condition that
- **for a single value of x it should have only a single value of y**

Thus a function is a mapping between input and output where multiple inputs can have same output but a single input cannot have multiple outputs

Equation of a circle is not a function if we look from the lense of how a function is defined rather from the Machine learning perspective we'll call that as a non linear boundary

12. Limits

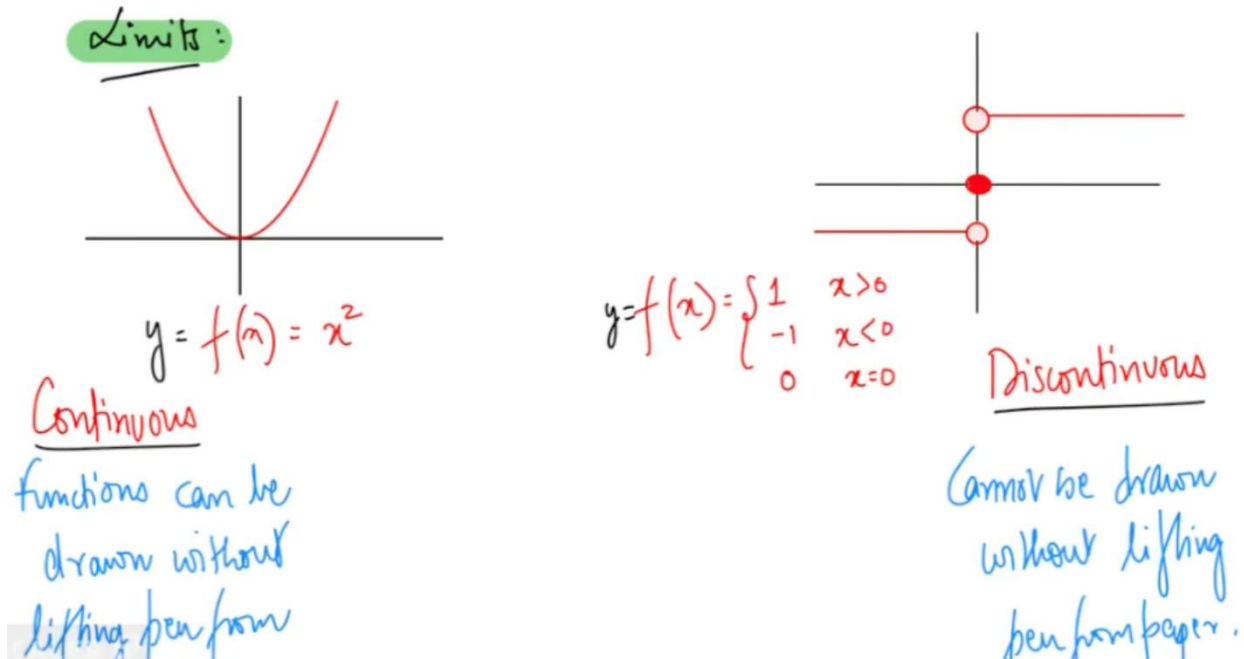
Limits

There are 2 types of functions

1. Continuous functions (left function)
 - A very basic defination of this function would be: they are function which **can** be draw wihtout lifting the pen from the paper

2. Discontinuous functions (Right function it is called as **signum function**)

- A very basic definition of this function would be: they are function which **cannot** be draw without lifting the pen from the paper



Now let's understand what is limits

Let's use our discontinuous function we'll learn about 2 new terminologies

1. Right hand limit

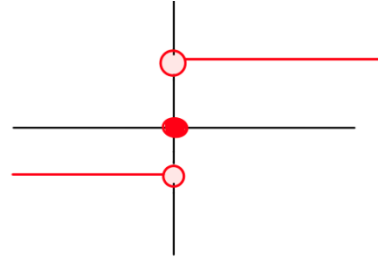
- It means we are approaching a value let say 0 from positive direction
- for the signum function we can see that as it is approaching 0 from the positive side the value tends to 1

2. Left hand limit

- It means we are approaching a value let say 0 from negative direction
- for the signum function we can see that as it is approaching 0 from the negative side the value tends to -1

Continuity:

$$y=f(x)=\begin{cases} 1 & x>0 \\ -1 & x<0 \\ 0 & x=0 \end{cases}$$



① Right Hand Limit

$$\lim_{x \rightarrow 0^+} f(x) = 1$$

$$x \rightarrow 0.1, 0.01, 0.001, 0.0001, 0.00001$$

② Left hand limit -

$$\lim_{x \rightarrow 0^-} f(x) = -1$$

$$x \rightarrow -0.1, -0.001, -0.00000001$$

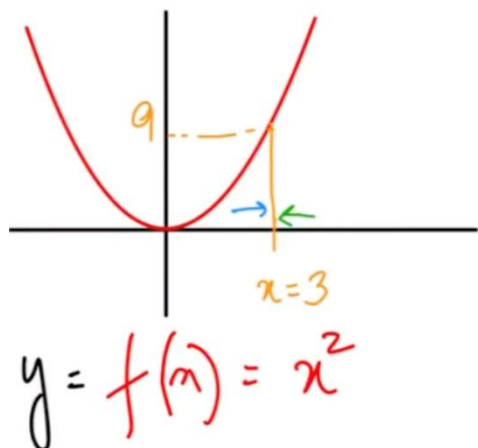
Let's look at continuous function now, and we pick any random value of x let say $x=3$ the corresponding value of y would be 9 since the function is $f(x)=x^2$

1. Right hand limit

- Here it means that what is the value of y when we are approaching the value of $x=3$ from the positive side ie from value of x greater than 3
- So as we reach very very close to the point of $x=3$ the y value tends to or becomes equal to 9
 - let's say that the x value is 3.0000001 can you see the corresponding y value be 9.0000006 which is very close to 9
 - thus we say that as we approach $x=3$ the value of y tends to 9

2. Left hand limit

- similarly here it means that what is the value of y when we are approaching the value of $x=3$ from the negative side ie from value of x less than 3
- so for value of $x=2.9999999$ the corresponding value of y would be 8.9999994 can you see it is very close to 9
- thus we say that as we approach $x=3$ from the left hand side the corresponding y value tends to 9



RHL

$$\lim_{x \rightarrow 3^+} f(x) = 9$$

$$\lim_{x \rightarrow 3^+} x^2 = 9$$

LHL

$$\lim_{x \rightarrow 3^-} f(x) = 9$$

$$\lim_{x \rightarrow 3^-} x^2 = 9$$

We call a function as continuous if it satisfies the below condition

- left hand limit is equal to right hand limit is equal to the $f(x)$ value at $x=a$
- if we come from either of the side we should reach the same point
- continuity is measured at a specific point
- continuous function is one that at all valid x values it is continuous

Function is Continuous if

$$LHL = RHL = f(x)_{x=a}$$

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$$

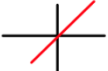
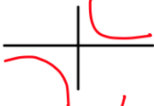
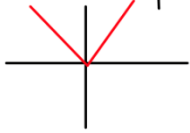
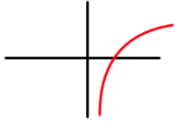
Some important functions

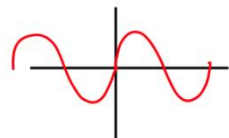
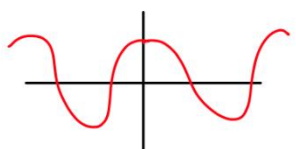
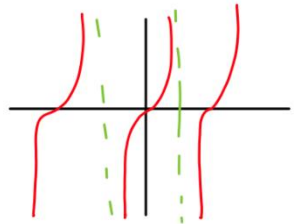
Instructors note:

- The instructor can take each function demonstrate its plot in desmos and discuss about the domain, range and whether it is continuous or discontinuous based on the concepts studied

Let's learn about some important functions that you'll be seeing throughout your ML and DL journey

Some Important Functions:

Functions	Domain	Range	Continuous	Plot
① $y = x$	$(-\infty, \infty)$	$(-\infty, \infty)$	Continuous	
② $y = \frac{1}{x}$	$(-\infty, \infty)$	$(-\infty, \infty)$	discontinuous	
③ $y = e^x$	$(-\infty, \infty)$	$(0, \infty)$	continuous	
④ $y = x $	$(-\infty, \infty)$	$(0, \infty)$	continuous	
⑤ $y = \ln(x)$	$(0, \infty)$	$(-\infty, \infty)$	continuous	

⑥	$y = \frac{1}{1+e^{-x}} \quad (-\infty, \infty) \quad (0, 1)$	continuous	
⑦	$y = \sin \theta \quad (-\infty, \infty) \quad (-1, 1)$	continuous	
⑧	$y = \cos \theta \quad (-\infty, \infty) \quad [-1, 1]$	continuous	
⑨	$y = \tan \theta \quad (-\infty, \infty) \quad (-\infty, \infty)$	discontinuous	

$$g(\bar{x}_i, y_i, \bar{w}, w_0) = \left(\frac{\bar{w}^T \bar{x}_i + w_0}{\|\bar{w}\|} \right) y_i$$

$\bar{w}^*, w_0^* \rightarrow$ optimal values for best classifier

$$\bar{w}^*, w_0^* = \underset{\bar{w}, w_0}{\operatorname{argmax}} g(D, \bar{w}, w_0)$$

The best possible classifier is defined by $\bar{w}^*, w_0^*$, which will give me maximum value of gain function.